

i) Background

- 1-4 August 2023 was exceptionally smoky across the U.S. Midwest due to Canadian fires
- How does the Unified Forecast System with Chemistry (UFS-Chem) simulate meteorological variables during this event?**
- This study presents an evaluation case study using MELODIES-MONET (**M**odel **E**valuation using **O**bservations, **D**iagnostics and **E**xperiments **S**oftware – **M**odel and **O**bservation **E**valuation **T**oolkit), a Python-based model evaluation toolkit

EPA Region 5: AirNow overlaid on UFS-Chem

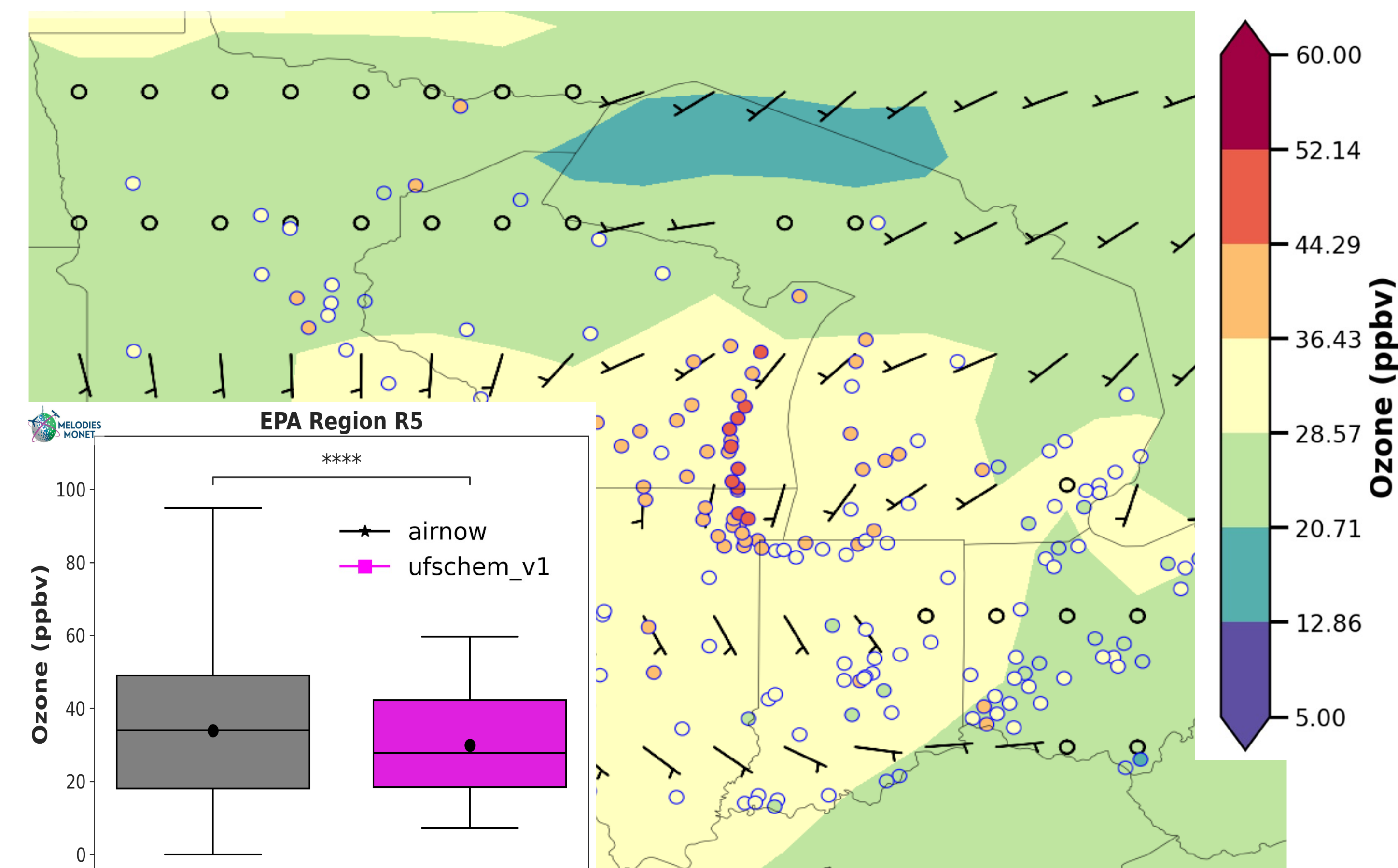


Fig. 2: Spatial overlay of AirNow on UFS-Chem. Fig. 2 shows that UFS-Chem significantly underestimated ozone pollution north of the Chicago metropolitan area.

iii) Timeseries and Vertical Validation

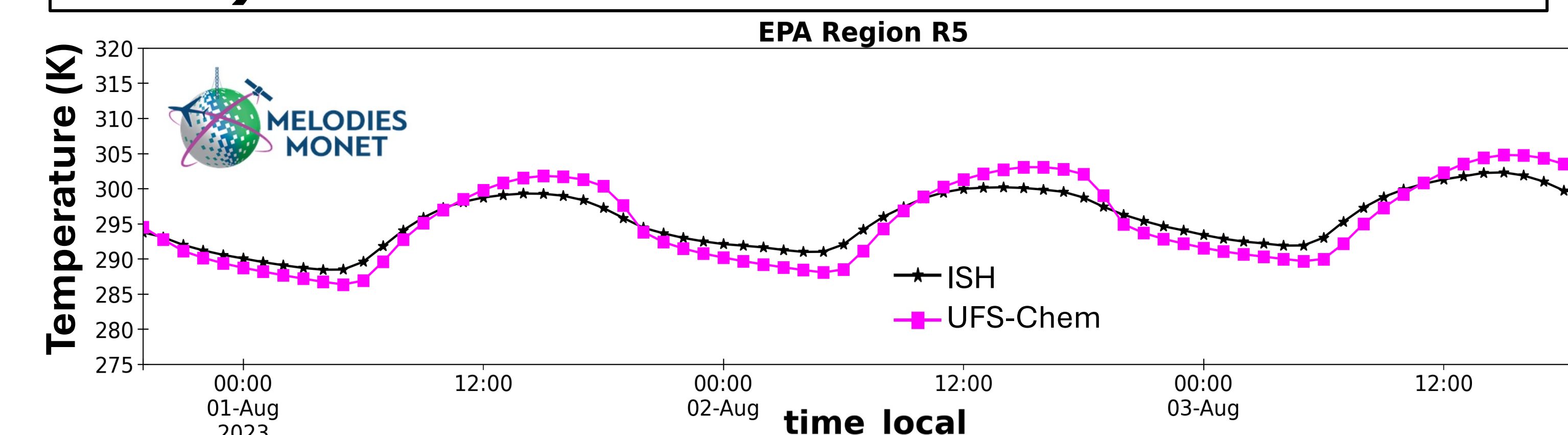


Fig. 4: Timeseries of UFS-Chem and ISH across EPA region 5. UFS-Chem slightly overestimates daily max temperature

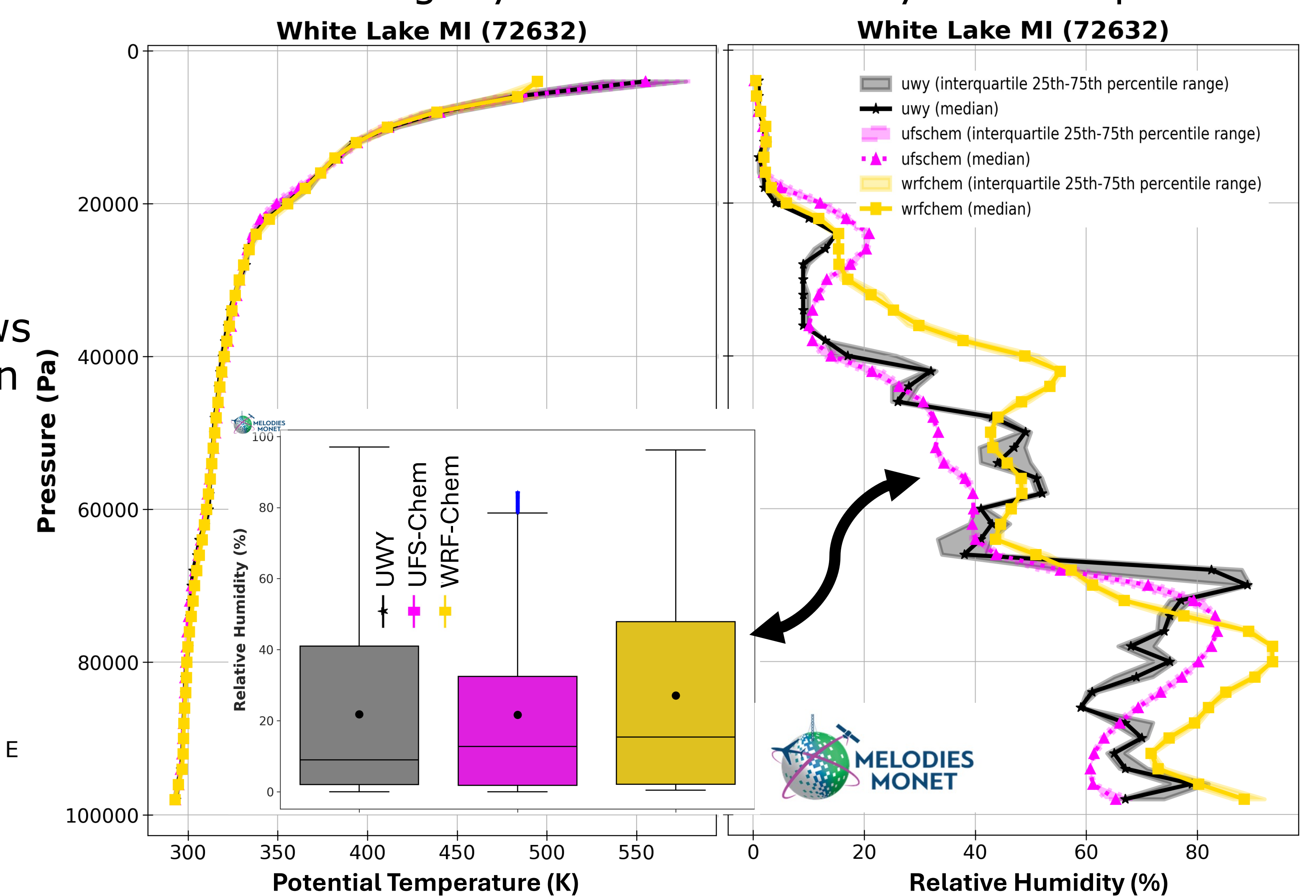
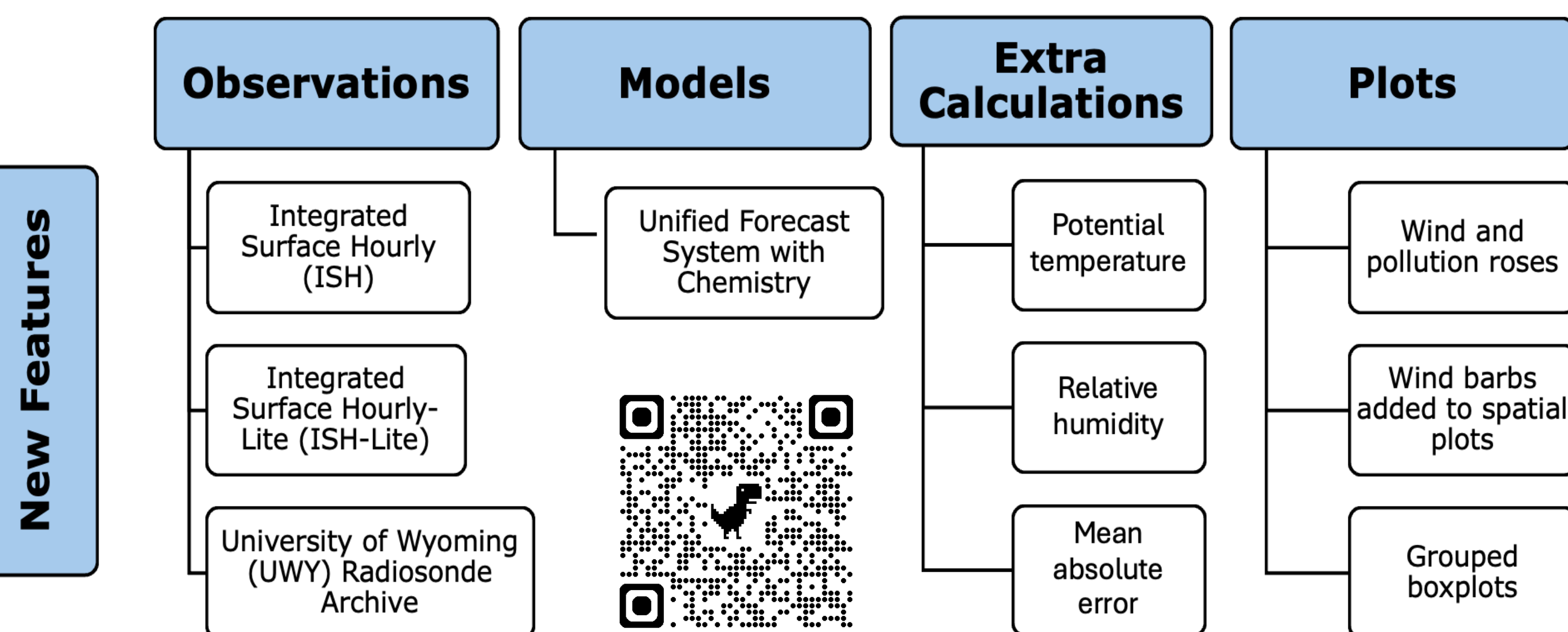


Fig. 5: UFS-Chem, validated against WRF-Chem, accurately simulates potential temperature throughout the atmosphere, but struggles to capture relative humidity to varying degrees



Schematic overview of the meteorological evaluation capabilities MELODIES-MONET users can now leverage

ii) 1-4 August 2023 Example Use Case

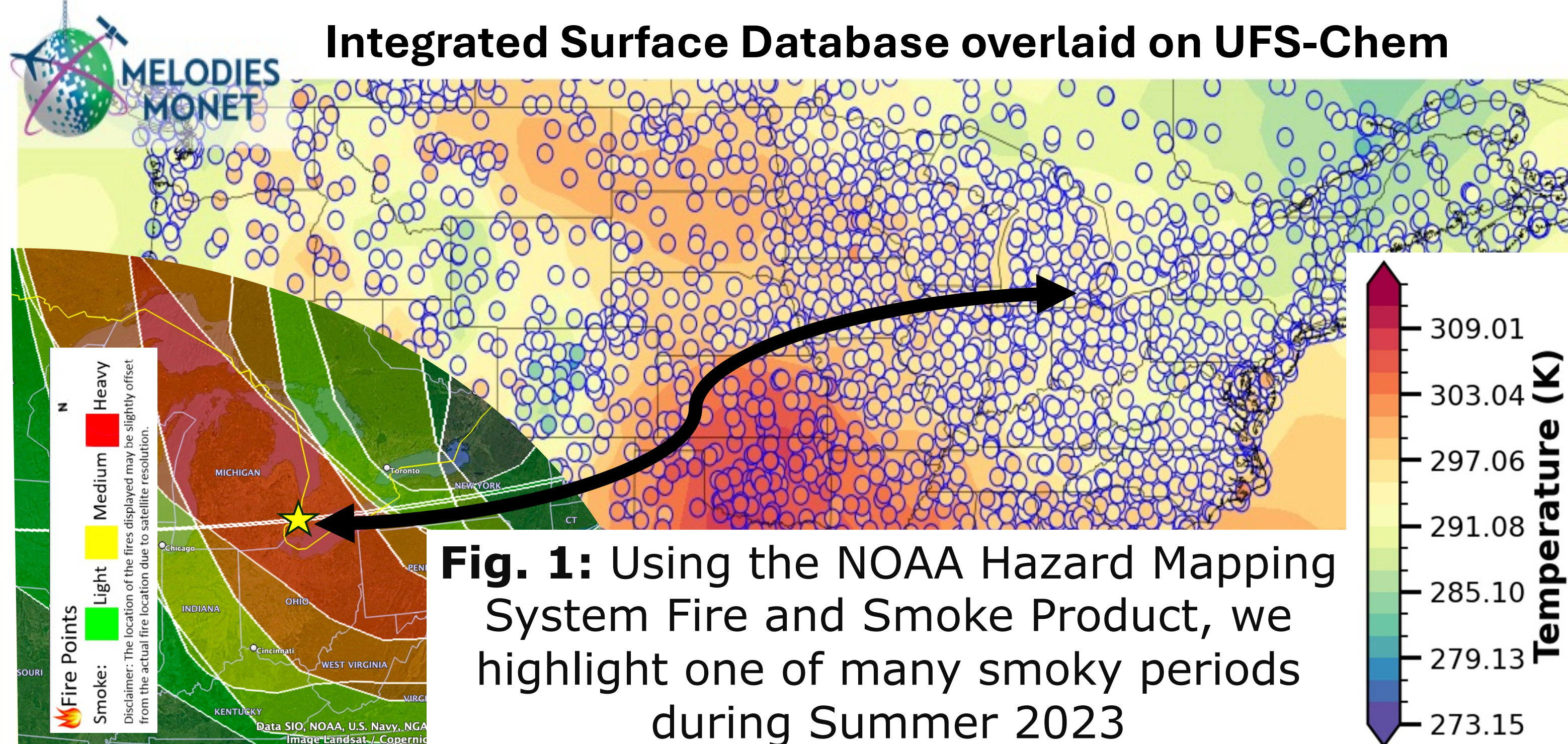


Fig. 1: Using the NOAA Hazard Mapping System Fire and Smoke Product, we highlight one of many smoky periods during Summer 2023

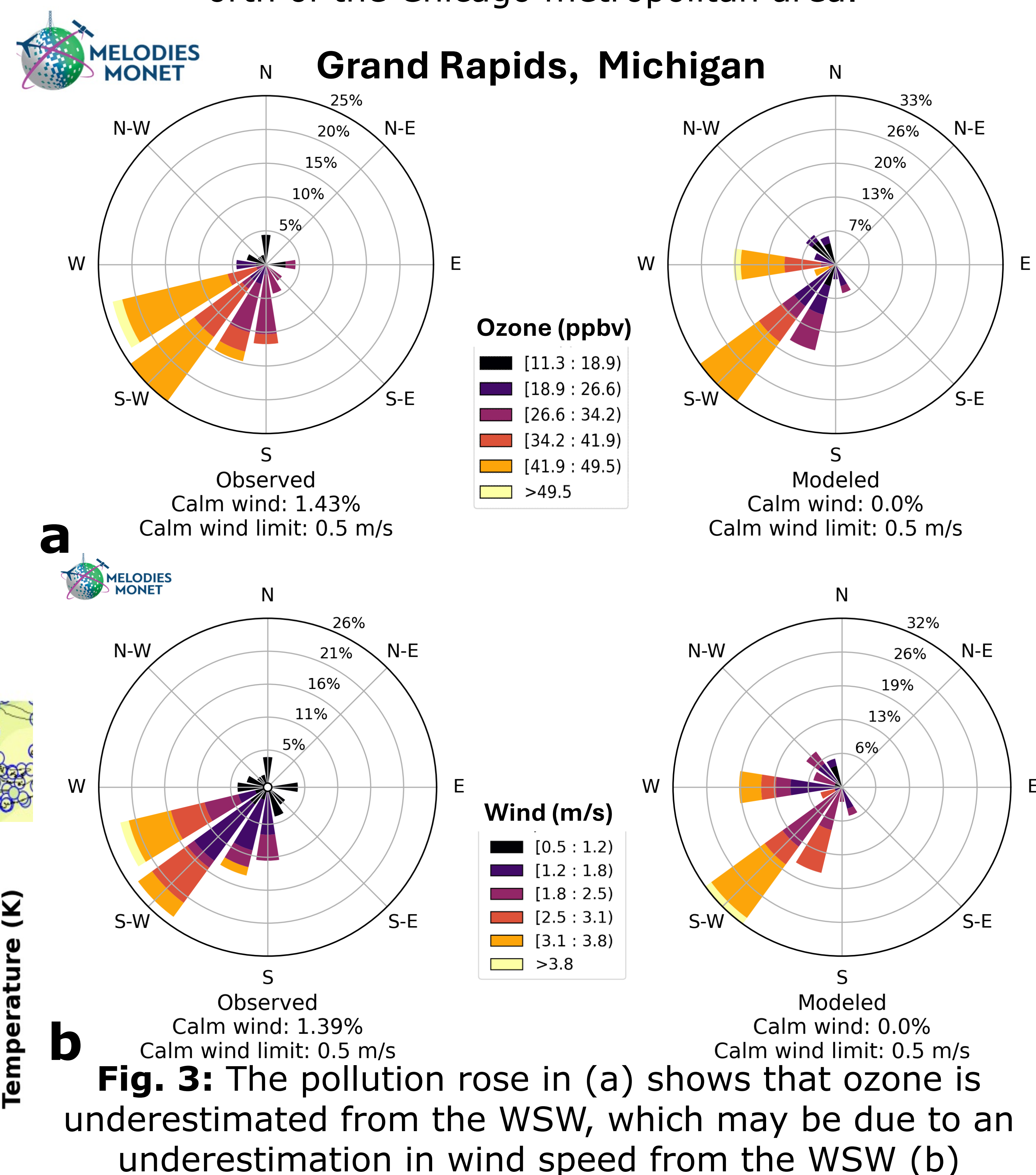


Fig. 3: The pollution rose in (a) shows that ozone is underestimated from the WSW, which may be due to an underestimation in wind speed from the WSW (b)

iv) Implications

- Inaccurate meteorology **may cause or exacerbate** biases in atmospheric chemistry
- MELODIES-MONET enables users to perform quick model evaluations to determine the role meteorology could have in driving biases within their atmospheric chemistry models

v) Acknowledgements

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